Учётная запись: `adm_ivanov`
Пароль: `Adm_P@ssw0rd12345`
Команда подключения:

bash

ssh adm_ivanov@<IP-адрес>

Хост 1: Роутер / Межсетевой экран (p-firewall-msk-01)

IP: `192.168.60.2`

1. Проверка парольной политики

- Проверить алгоритм хэширования:
  `grep ENCRYPT_METHOD /etc/login.defs` → ожидается `SHA512`.

- Проверить срок действия и длину:
  `grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs`

  `/etc/pam.d$ cat base-password`

  `# basic PAM configuration for Alpine.`

  `password required pam_unix.so nullok sha512 shadow`

  `# desktop keyrings`

  `-password optional pam_gnome_keyring.so use_authtok`

  `-password optional pam_kwallet5.so`

Ожидается: `PASS_MAX_DAYS 365`, `PASS_MIN_LEN 17` (или 16 для админов).

2. Запрет общих учётных записей

- `cat /etc/passwd | grep -E "admin|user|test|shared"` – не должно быть общих аккаунтов.

cat passwd

root:x:0:0:root:/root:/bin/sh

bin:x:1:1:bin:/bin:/sbin/nologin

daemon:x:2:2:daemon:/sbin:/sbin/nologin

lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin

sync:x:5:0:sync:/sbin:/bin/sync

shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown

halt:x:7:0:halt:/sbin:/sbin/halt

mail:x:8:12:mail:/var/mail:/sbin/nologin

news:x:9:13:news:/usr/lib/news:/sbin/nologin

uucp:x:10:14:uucp:/var/spool/uucppublic:/sbin/nologin

cron:x:16:16:cron:/var/spool/cron:/sbin/nologin

ftp:x:21:21::/var/lib/ftp:/sbin/nologin

sshd:x:22:22:sshd:/dev/null:/sbin/nologin

games:x:35:35:games:/usr/games:/sbin/nologin

ntp:x:123:123:NTP:/var/empty:/sbin/nologin

guest:x:405:100:guest:/dev/null:/sbin/nologin

nobody:x:65534:65534:nobody:/:/sbin/nologin

adm_ivanov:x:1000:1001::/home/adm_ivanov:/bin/bash

- `last root` – посмотреть, не использовался ли root для входа.

last root

BusyBox v1.36.1 (2025-11-23 14:32:18 UTC) multi-call binary.

Usage: last

Show listing of the last users that logged into the system

3. Принцип минимальных привилегий

- `getent group sudo` или `wheel` – проверить, что в группе только необходимые пользователи (например, `adm_ivanov`).

p-firewall-msk-01:/etc$ getent group admins

admins:x:1000:adm_ivanov

p-firewall-msk-01:/etc$ groups

adm_ivanov admins

- Проверить `/etc/sudoers` на наличие избыточных правил.

sudoers file.

##

This file MUST be edited with the 'visudo' command as root.

Failure to use 'visudo' may result in syntax or file permission errors

that prevent sudo from running.

##

See the sudoers man page for the details on how to write a sudoers file.

##

##

Host alias specification

##

Groups of machines. These may include host names (optionally with wildcards),

IP addresses, network numbers or netgroups.

Host_Alias   WEBSERVERS = www1, www2, www3

```
##
## User alias specification
##
## Groups of users.  These may consist of user names, uids, Unix groups,
## or netgroups.
# User_Alias    ADMINS = millert, dowdy, mikef


##
## Cmnd alias specification
##
## Groups of commands.  Often used to group related commands together.
# Cmnd_Alias    PROCESSES = /usr/bin/nice, /bin/kill, /usr/bin/renice, \
#                 /usr/bin/pkill, /usr/bin/top
#
# Cmnd_Alias    REBOOT = /sbin/halt, /sbin/reboot, /sbin/poweroff
#
# Cmnd_Alias    DEBUGGERS = /usr/bin/gdb, /usr/bin/lldb, /usr/bin/strace, \
#                 /usr/bin/truss, /usr/bin/bpftrace, \
#                 /usr/bin/dtrace, /usr/bin/dtruss
#
# Cmnd_Alias    PKGMAN = /usr/bin/apt, /usr/bin/dpkg, /usr/bin/rpm, \
#               /usr/bin/yum, /usr/bin/dnf,  /usr/bin/zypper, \
#               /usr/bin/pacman


##
## Defaults specification
##
## You may wish to keep some of the following environment variables
```

```
## when running commands via sudo.

##

## Locale settings

# Defaults env_keep += "LANG LANGUAGE LINGUAS LC_* _XKB_CHARSET"

##

## Run X applications through sudo; HOME is used to find the

## .Xauthority file.  Note that other programs use HOME to find

## configuration files and this may lead to privilege escalation!

# Defaults env_keep += "HOME"

##

## X11 resource path settings

# Defaults env_keep += "XAPPLRESDIR XFILESEARCHPATH
XUSERFILESEARCHPATH"

##

## Desktop path settings

# Defaults env_keep += "QTDIR KDEDIR"

##

## Allow sudo-run commands to inherit the callers' ConsoleKit session

# Defaults env_keep += "XDG_SESSION_COOKIE"

##

## Uncomment to enable special input methods.  Care should be taken as

## this may allow users to subvert the command being run via sudo.

# Defaults env_keep += "XMODIFIERS GTK_IM_MODULE QT_IM_MODULE
QT_IM_SWITCHER"

##

## Uncomment to use a hard-coded PATH instead of the user's to find commands

# Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"

##

## Uncomment to restore the historic behavior where a command is run in
```

```
## the user's own terminal.
# Defaults !use_pty
##
## Uncomment to send mail if the user does not enter the correct password.
# Defaults mail_badpass
##
## Uncomment to enable logging of a command's output, except for
## sudoreplay and reboot.  Use sudoreplay to play back logged sessions.
## Sudo will create up to 2,176,782,336 I/O logs before recycling them.
## Set maxseq to a smaller number if you don't have unlimited disk space.
# Defaults log_output
# Defaults!/usr/bin/sudoreplay !log_output
# Defaults!/usr/local/bin/sudoreplay !log_output
# Defaults!REBOOT !log_output
# Defaults maxseq = 1000
##
## Uncomment to disable intercept and log_subcmds for debuggers and
## tracers.  Otherwise, anything that uses ptrace(2) will be unable
## to run under sudo if intercept_type is set to "trace".
# Defaults!DEBUGGERS !intercept, !log_subcmds
##
## Uncomment to disable intercept and log_subcmds for package managers.
## Some package scripts run a huge number of commands, which is made
## slower by these options and also can clutter up the logs.
# Defaults!PKGMAN !intercept, !log_subcmds


##
## Runas alias specification
```

```
##


##
## User privilege specification
##
root ALL=(ALL:ALL) ALL


## Uncomment to allow members of group wheel to execute any command
# %wheel ALL=(ALL:ALL) ALL


## Same thing without a password
# %wheel ALL=(ALL:ALL) NOPASSWD: ALL


## Uncomment to allow members of group sudo to execute any command
# %sudo    ALL=(ALL:ALL) ALL


## Uncomment to allow any user to run sudo if they know the password
## of the user they are running the command as (root by default).
# Defaults targetpw  # Ask for the password of the target user
# ALL ALL=(ALL:ALL) ALL  # WARNING: only use this together with 'Defaults targetpw'


## Read drop-in files from /etc/sudoers.d
@includedir /etc/sudoers.d
Defaults    logfile="/var/log/sudo.log"
Defaults    log_input, log_output
```

4. Логирование sudo и контроль администраторов

- `grep "logfile" /etc/sudoers` – должна быть строка `Defaults logfile="/var/log/sudo.log"`.

  p-firewall-msk-01:/etc$ sudo grep "logfile" /etc/sudoers

  Defaults   logfile="/var/log/sudo.log"

- `ls -la /var/log/sudo.log` – права 640, владелец root:adm.

p-firewall-msk-01:/etc$ ls -la /var/log/sudo.log

-rw------- 1 root root 323 Aug 29 12:36 /var/log/sudo.log

5. Запрет root login по SSH

- `grep -i permitrootlogin /etc/ssh/sshd_config` → должно быть `PermitRootLogin no`.

p-firewall-msk-01:/etc$ grep -i permitrootlogin /etc/ssh/sshd_config

PermitRootLogin no

the setting of "PermitRootLogin prohibit-password".

- `sshd -T | grep permitrootlogin` – подтвердить.

6. Сетевая безопасность (файрвол, политика deny by default)

- `iptables -L -n -v` – проверить, что политики `INPUT`, `FORWARD`, `OUTPUT` имеют `DROP` или `REJECT` по умолчанию.

p-firewall-msk-01:/etc$ sudo iptables -L -n -v

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)

 pkts bytes target prot opt in out source destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)

 pkts bytes target prot opt in out source destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)

 pkts bytes target prot opt in out source destination

- Проверить, разрешены ли только необходимые порты (SSH, HTTP/HTTPS, DNS).
- Если используется `firewalld`: `firewall-cmd --list-all`. не ипользуется

7. Отключение неиспользуемых сервисов

- `systemctl list-unit-files --state=enabled` – проверить, нет ли лишних служб (например, `named`, `httpd`).

8. Настройка sysctl (стандарт Linux)

- Проверить файл `/etc/sysctl.conf` на наличие параметров:
  `net.ipv4.conf.default.send_redirects = 0`,
  `net.ipv4.conf.all.accept_redirects = 0`,
  `net.ipv4.conf.default.accept_redirects = 0`,
  `net.ipv4.conf.all.secure_redirects = 0`,
  `net.ipv4.conf.default.secure_redirects = 0`,
  `net.ipv4.conf.all.accept_source_route = 0`,
  `net.ipv4.conf.default.accept_source_route = 0`,
  `net.ipv4.tcp_synookies = 1`.

cat rsyslog.conf

rsyslog configuration file

#

For more information see /usr/share/doc/rsyslog-*/rsyslog_conf.html

or latest version online at http://www.rsyslog.com/doc/rsyslog_conf.html

If you experience problems, see http://www.rsyslog.com/doc/troubleshoot.html

Global directives

Sets the directory that rsyslog uses for work files.

$WorkDirectory /var/lib/rsyslog

Sets default permissions for all log files.

$FileOwner root

$FileGroup adm

```
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022

# Check config syntax on startup and abort if unclean (default off).
#$AbortOnUncleanConfig on

# Reduce repeating messages (default off).
#$RepeatedMsgReduction on



#### Modules ####

# Provides --MARK-- message capability.
module(load="immark")

# Provides support for local system logging (e.g. via logger command).
module(load="imuxsock")

# Reads kernel messages.
module(load="imklog")

#### Config files ####

# Include all config files in /etc/rsyslog.d/.
include(file="/etc/rsyslog.d/*.conf" mode="optional")

#### Rules ####
```

```
# Log all kernel messages to kern.log.
kern.*                          /var/log/kern.log

# Log anything (except mail) of level info or higher.
# Don't log private authentication messages!
# NOTE: The minus sign in front of filename disables buffer flush.
*.info;authpriv.none;cron.none;kern.none;mail.none     -/var/log/messages

# The authpriv file has restricted access.
authpriv.*                       /var/log/auth.log

# Log all the mail messages in one place.
mail.*                          -/var/log/mail.log

# Log cron stuff.
cron.*                          -/var/log/cron.log

# Everybody gets emergency messages.
*.emerg                          :omusrmsg:*

# Log all kernel messages to the console.
# Logging much else clutters up the screen.
#kern.*                          /dev/console


### Examples ####
```

```
# Send all logs to remote syslog via UDP.
# An on-disk queue is created for this action. If the remote host is
# down, messages are spooled to disk and sent when it is up again.
#*.* action(
#    type="omfwd"
#    target="192.168.0.1"
#    port="514"
#    protocol="udp"
#    queue.filename="fwdRule1"  # unique name prefix for spool files
#    queue.type="LinkedList"
#    queue.maxDiskSpace="256m"
#    queue.saveOnShutdown="on"
#    action.resumeRetryCount="-1"
#    action.resumeInterval="30"
#)

# Receive messages from remote host via UDP
# for parameters see http://www.rsyslog.com/doc/imudp.html
#module(load="imudp")  # needs to be done just once
#input(
#    type="imudp"
#    port="514"
#)
```

- Применить: `sysctl -p`, проверить значения: `sysctl net.ipv4.tcp_synookies` и т.д.

9. Ограничение su (только группа admins)

- Проверить `/etc/pam.d/su` – должна быть раскомментирована строка:

  # basic PAM configuration for Alpine.

  auth sufficient pam_rootok.so

  auth include base-auth

  account include base-account

  password include base-password

  session include base-session-noninteractive

  p-firewall-msk-01:/etc/pam.d$ cat base-auth

  # basic PAM configuration for Alpine.

  auth required pam_unix.so nullok

  auth required pam_nologin.so

  auth required pam_env.so

  -auth optional pam_gnome_keyring.so

  -auth optional pam_kwallet5.so

  p-firewall-msk-01:/etc/pam.d$ cat base-account

  # basic PAM configuration for Alpine.

  account required pam_unix.so

  account required pam_nologin.so

  p-firewall-msk-01:/etc/pam.d$ cat base-password

  # basic PAM configuration for Alpine.

  password required pam_unix.so nullok sha512 shadow

  # desktop keyrings

-password optional pam_gnome_keyring.so use_authtok

-password optional pam_kwallet5.so

p-firewall-msk-01:/etc/pam.d$ cat base-session

basic PAM configuration for Alpine.

session include base-session-noninteractive

potential providers that set up session stuff / a XDG_RUNTIME_DIR

-session optional pam_rundir.so

-session optional pam_elogind.so

-session optional pam_ck_connector.so

-session optional pam_turnstile.so

-session optional pam_dumb_runtime_dir.so

desktop keyrings

-session optional pam_gnome_keyring.so auto_start

-session optional pam_kwallet5.so auto_start

p-firewall-msk-01:/etc/pam.d$ cat base-session-noninteractive

basic PAM configuration for Alpine.

session required pam_env.so

session required pam_limits.so

session required pam_unix.so

`auth required pam_wheel.so use_uid group=admins`.

10. Логирование событий безопасности

- `systemctl status rsyslog` – должна быть запущена.
- `ls -la /var/log/auth.log` (или secure) – права 640, root:adm.

p-firewall-msk-01:/etc/pam.d$ ls -la /var/log/auth.log

-rw-r--r-- 1 root root 453 Aug 29 12:37 /var/log/auth.log

- `ls -la /var/log/syslog` (или messages) – аналогично.

p-firewall-msk-01:/etc/pam.d$ ls -la /var/log/syslog

-rw-r--r-- 1 root root 0 Aug 29 12:21 /var/log/syslog

11. Проверка именования

- `hostname` – должно быть `p-firewall-msk-01` (соответствует стандарту: `prod-firewall-msk-01`? Здесь `p` вероятно `prod`, формат соблюдён).

p-firewall-msk-01:/etc/pam.d$ hostname

p-firewall-msk-01

Хост 2: Веб-сервер (p-nginx-msk-02)

IP: `192.168.10.10`

Выполнить **все те же проверки**, что и для роутера (пункты 1–5, 7–11), **плюс специфичные для Nginx**:

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs

PASS_MAX_DAYS Maximum number of days a password may be used.

PASS_WARN_AGE Number of days warning given before a password expires.

PASS_MAX_DAYS 99999

PASS_WARN_AGE 7

#PASS_MIN_LEN

```
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ cat /etc/passwd | grep -E "admin|user|test|
shared"
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
test:x:1001:1002::/home/test:/bin/bash
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ last root

wtmp begins Sat Aug 29 12:44:01 2026

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ cat /etc/sudoers | grep -v "^#" | grep -v "^$"
cat: /etc/sudoers: Permission denied
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"
Defaults    env_reset
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults    use_pty
root    ALL=(ALL:ALL) ALL
%admin ALL=(ALL) ALL
%sudo    ALL=(ALL:ALL) ALL
@includedir /etc/sudoers.d
Defaults    logfile="/var/log/sudo.log"
Defaults    log_input, log_output

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ sudo grep logfile /etc/sudoers
Defaults    logfile="/var/log/sudo.log"
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ ls -la /var/log/sudo.log
-rw------- 1 root root 336 Aug 29 12:55 /var/log/sudo.log
```

```
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ grep -i permitrootlogin /etc/ssh/sshd_config
PermitRootLogin no
# the setting of "PermitRootLogin without-password".

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ iptables -L -n -v
-bash: iptables: command not found
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ ufw status
-bash: ufw: command not found
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ firewall-cmd --list-all
-bash: firewall-cmd: command not found
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ systemctl list-unit-files --state=enabled
-bash: systemctl: command not found

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ grep -E "send_redirects|accept_redirects|
secure_redirects|accept_source_route|tcp_synookies" /etc/sysctl.conf
net.ipv4.conf.all.accept_redirects=0
#net.ipv6.conf.all.accept_redirects = 0
net.ipv4.conf.all.secure_redirects=0
#net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.all.accept_source_route=0
#net.ipv6.conf.all.accept_source_route = 0
net.ipv4.conf.default.send_redirects=0
net.ipv4.conf.default.accept_redirects=0
net.ipv4.conf.default.secure_redirects=0
net.ipv4.conf.default.accept_source_route=0
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ sysctl net.ipv4.tcp_synookies
sysctl: cannot stat /proc/sys/net/ipv4/tcp_synookies: No such file or directory
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0
```

```
adm_ivanov@p-nginx-msk-02:/etc/pam.d$ grep pam_wheel /etc/pam.d/su
# auth       required   pam_wheel.so
# auth       sufficient pam_wheel.so trust
# auth       required   pam_wheel.so deny group=nosu

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ ls -la /var/log/
total 388
drwxr-xr-x 1 root root   4096 Aug 29 12:21 .
drwxr-xr-x 1 root root   4096 Aug 29 12:19 ..
-rw-r--r-- 1 root root  10344 Aug 29 12:20 alternatives.log
drwxr-xr-x 1 root root   4096 Aug 29 12:19 apt
-rw-r--r-- 1 root root  64549 Aug 10 14:50 bootstrap.log
-rw-rw---- 1 root utmp      0 Aug 10 14:50 btmp
-rw-r--r-- 1 root root 227933 Aug 29 12:20 dpkg.log
-rw-r--r-- 1 root root  32064 Aug 29 12:21 faillog
-rw-rw-r-- 1 root utmp 292584 Aug 29 12:44 lastlog
drwxr-xr-x 1 root adm    4096 Aug 29 12:19 nginx
drwx------ 3 root root   4096 Aug 29 12:21 sudo-io
-rw------- 1 root root    336 Aug 29 12:55 sudo.log
-rw-rw-r-- 1 root utmp    384 Aug 29 12:44 wtmp

adm_ivanov@p-nginx-msk-02:/etc/pam.d$ hostname
p-nginx-msk-02
```

12. Проверка конфигурации Nginx

- Открыть файл `/etc/nginx/nginx.conf` и/или в `/etc/nginx/conf.d/`.
- Проверить наличие:
  - `server_tokens off;`
  - `client_max_body_size 10m;`
  - `client_body_timeout 30s;`
  - `send_timeout 10s;`
  - `keepalive_timeout 65;`
  - Если используется HTTPS:
    `ssl_protocols TLSv1.2 TLSv1.3;`
    `ssl_ciphers ECDHE-RSA-AES256-GCM-SHA512:DHE-RSA-AES256-GCM-SHA512:...;` (указанные в стандарте)
  - Заголовки безопасности:
    `add_header X-Frame-Options "SAMEORIGIN" always;`
    `add_header X-Content-Type-Options "nosniff" always;`
    `add_header Referrer-Policy "strict-origin-when-cross-origin" always;`
    `add_header Content-Security-Policy "default-src 'self';" always;`
- Проверить, что процесс Nginx запущен от непривилегированного пользователя: `ps aux | grep nginx` – мастер-процесс должен быть от `nginx` или `www-data`, не от root.

adm_ivanov@p-nginx-msk-02:/etc/nginx/http.d$ ps aux | grep nginx

root 85 0.0 0.0 12120 6396 ? S 12:21 0:00 nginx: master process nginx -g daemon off;

root 87 0.0 0.0 12672 2860 ? S 12:21 0:00 nginx: worker process

root 88 0.0 0.0 12672 2860 ? S 12:21 0:00 nginx: worker process

root 89 0.0 0.0 12672 2860 ? S 12:21 0:00 nginx: worker process

root 90 0.0 0.0 12672 2732 ? S 12:21 0:00 nginx: worker process

adm_iva+ 113 0.0 0.0 4024 2048 pts/0 S+ 12:45 0:00 grep --color=auto nginx

- Проверить наличие логов: `ls -la /var/log/nginx/`. adm_ivanov@p-nginx-msk-02:/etc/nginx/http.d$ ls -la /var/log/nginx/
- total 16
- drwxr-xr-x 1 root adm 4096 Aug 29 12:19 .
- drwxr-xr-x 1 root root 4096 Aug 29 12:21 ..

- -rw-r----- 1 www-data adm 0 Aug 29 12:19 access.log
- -rw-r----- 1 www-data adm 0 Aug 29 12:19 error.log
-

adm_ivanov@p-nginx-msk-02:~$ cat /etc/nginx/nginx.conf

/etc/nginx/nginx.conf

user root;

Set number of worker processes automatically based on number of CPU cores.

worker_processes auto;

Enables the use of JIT for regular expressions to speed-up their processing.

pcre_jit on;

pid /var/run/nginx.pid;

Configures default error logger.

error_log /var/log/nginx/error.log warn;

Includes files with directives to load dynamic modules.

include /etc/nginx/modules/*.conf;

Include files with config snippets into the root context.

include /etc/nginx/conf.d/*.conf;

events {

    # The maximum number of simultaneous connections that can be opened by

    # a worker process.

```
    worker_connections 1024;
}

http {
    # Includes mapping of file name extensions to MIME types of responses
    # and defines the default type.
    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    # Name servers used to resolve names of upstream servers into addresses.
    # It's also needed when using tcpsocket and udpsocket in Lua modules.
    #resolver 1.1.1.1 1.0.0.1 [2606:4700:4700::1111] [2606:4700:4700::1001];

    # Don't tell nginx version to the clients. Default is 'on'.
    # server_tokens off;

    # Specifies the maximum accepted body size of a client request, as
    # indicated by the request header Content-Length. If the stated content
    # length is greater than this size, then the client receives the HTTP
    # error code 413. Set to 0 to disable. Default is '1m'.
    client_max_body_size 100m;

    # Sendfile copies data between one FD and other from within the kernel,
    # which is more efficient than read() + write(). Default is off.
    sendfile on;

    # Causes nginx to attempt to send its HTTP response head in one packet,
    # instead of using partial frames. Default is 'off'.
```

```
    tcp_nopush on;



    # Enables the specified protocols. Default is TLSv1 TLSv1.1 TLSv1.2.
    # TIP: If you're not obligated to support ancient clients, remove TLSv1.1.
    ssl_protocols TLSv1.1 TLSv1.2 TLSv1.3;

    # Path of the file with Diffie-Hellman parameters for EDH ciphers.
    # TIP: Generate with: `openssl dhparam -out /etc/ssl/nginx/dh2048.pem 2048`
    #ssl_dhparam /etc/ssl/nginx/dh2048.pem;

    # Specifies that our cipher suits should be preferred over client ciphers.
    # Default is 'off'.
    ssl_prefer_server_ciphers on;
    ssl_ciphers ECDHE-RSA-AES256-GCM-SHA512:DHE-RSA-AES256-GCM-
SHA512:ECDHE-RSA-AES256-GCM-SHA384:DHE-RSA-AES256-GCM-SHA384;

    # Enables a shared SSL cache with size that can hold around 8000 sessions.
    # Default is 'none'.
    ssl_session_cache shared:SSL:2m;

    # Specifies a time during which a client may reuse the session parameters.
    # Default is '5m'.
    ssl_session_timeout 1d;

    # Disable TLS session tickets (they are insecure). Default is 'on'.
    ssl_session_tickets on;
```

```
# Enable gzipping of responses.
#gzip on;

# Set the Vary HTTP header as defined in the RFC 2616. Default is 'off'.
gzip_vary on;


add_header Referrer-Policy "strict-origin-when-cross-origin" always;
add_header Content-Security-Policy "default-src 'self';" always;


# Helper variable for proxying websockets.
map $http_upgrade $connection_upgrade {
    default upgrade;
    '' close;
}


# Specifies the main log format.
log_format main '$remote_addr - $remote_user [$time_local] "$request" '
        '$status $body_bytes_sent "$http_referer" '
        '"$http_user_agent" "$http_x_forwarded_for"';

# Sets the path, format, and configuration for a buffered log write.
access_log /var/log/nginx/access.log main;


# Includes virtual hosts configs.
```

```
    include /etc/nginx/http.d/*.conf;
}
```

```
adm_ivanov@p-nginx-msk-02:/etc/nginx/http.d$ cat zz-server.conf
server {
    listen 80 default_server;
    server_name site.local;

    root /usr/share/nginx/html;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
```

ХОСТ 3 – файловый сервер (p-fs-msk-03)

IP: `192.168.10.40`
Подключение: `ssh adm_ivanov@192.168.10.40`

Выполните следующие команды (аналогичные общим, плюс специфичные для файлового сервера):

Общие проверки (повторяем как для хоста 2)

1. **Парольная политика**
   bash

grep ENCRYPT_METHOD /etc/login.defs
grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs

Запрет общих учётных записей

bash
cat /etc/passwd | grep -E "admin|user|test|shared"
last root

Минимальные привилегии (группы sudo)

bash
getent group sudo
getent group wheel
sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"

Логирование sudo

bash
sudo grep logfile /etc/sudoers
ls -la /var/log/sudo.log

Запрет root login по SSH

bash
grep -i permitrootlogin /etc/ssh/sshd_config
sshd -T | grep permitrootlogin

Сетевая фильтрация

bash
iptables -L -n -v
если есть firewalld:
firewall-cmd --list-all
если ufw:
ufw status

Отключение неиспользуемых сервисов

bash
systemctl list-unit-files --state=enabled 2>/dev/null || echo "systemctl not available"
systemctl list-units --type=service --state=running 2>/dev/null || echo "systemctl not available"

Sysctl

bash

grep -E "send_redirects|accept_redirects|secure_redirects|accept_source_route|
tcp_synookies" /etc/sysctl.conf
sysctl net.ipv4.tcp_synookies
sysctl net.ipv4.conf.all.accept_redirects

Ограничение su

bash
grep pam_wheel /etc/pam.d/su

Логирование событий

bash
systemctl status rsyslog 2>/dev/null || echo "rsyslog not found"
ls -la /var/log/auth.log   # или secure
ls -la /var/log/syslog   # или messages

Именование

bash
hostname
hostnamectl

Специфичные для файлового сервера

12. **Проверка прав на каталоги** (из отчёта пентеста)
    bash

ls -la /srv/files   # или путь, где хранятся файлы ШАРЫ НЕТ
проверить владельца и права (должны быть root:fs_admins, 750)

Проверка избыточных административных прав

bash
посмотреть, какие пользователи имеют доступ к общим каталогам
getfacl /srv/files # если поддерживается
или проверить группы пользователей
groups adm_ivanov
и других пользователей, если есть

adm_ivanov@p-fs-msk-03:~$ grep ENCRYPT_METHOD /etc/login.defs
This variable is deprecated. You should use ENCRYPT_METHOD.
ENCRYPT_METHOD SHA512
Only used if ENCRYPT_METHOD is set to SHA256 or SHA512.
adm_ivanov@p-fs-msk-03:~$ grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|
PASS_WARN_AGE" /etc/login.defs
PASS_MAX_DAYS   Maximum number of days a password may be used.
PASS_WARN_AGE   Number of days warning given before a password expires.

```
PASS_MAX_DAYS 60
PASS_WARN_AGE 14
#PASS_MIN_LEN
PASS_MIN_LEN 17
adm_ivanov@p-fs-msk-03:~$ cat /etc/passwd | grep -E "admin|user|test|shared"
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
adm_ivanov@p-fs-msk-03:~$ last root

wtmp begins Sat Aug 29 13:00:04 2026
adm_ivanov@p-fs-msk-03:~$ getent group sudo
sudo:x:27:
adm_ivanov@p-fs-msk-03:~$ getent group wheel
adm_ivanov@p-fs-msk-03:~$ sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"
Defaults    env_reset
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults    use_pty
root    ALL=(ALL:ALL) ALL
%admin ALL=(ALL) ALL
%sudo    ALL=(ALL:ALL) ALL
@includedir /etc/sudoers.d
Defaults    logfile="/var/log/sudo.log"
Defaults    log_input, log_output
adm_ivanov@p-fs-msk-03:~$ sudo grep logfile /etc/sudoers
Defaults    logfile="/var/log/sudo.log"
adm_ivanov@p-fs-msk-03:~$ ls -la /var/log/sudo.log
-rw------- 1 root root 348 Aug 29 13:00 /var/log/sudo.log
adm_ivanov@p-fs-msk-03:~$ grep -i permitrootlogin /etc/ssh/sshd_config
PermitRootLogin no
# the setting of "PermitRootLogin without-password".
adm_ivanov@p-fs-msk-03:~$ sshd -T | grep permitrootlogin
sshd: no hostkeys available -- exiting.
adm_ivanov@p-fs-msk-03:~$ iptables -L -n -v
-bash: iptables: command not found
adm_ivanov@p-fs-msk-03:~$ firewall-cmd --list-all
-bash: firewall-cmd: command not found
adm_ivanov@p-fs-msk-03:~$ ufw status
-bash: ufw: command not found
adm_ivanov@p-fs-msk-03:~$ systemctl list-unit-files --state=enabled 2>/dev/null || echo
"systemctl not available"
systemctl not available
adm_ivanov@p-fs-msk-03:~$ systemctl list-units --type=service --state=running 2>/dev/null
|| echo "systemctl not available"
systemctl not available
adm_ivanov@p-fs-msk-03:~$ grep -E "send_redirects|accept_redirects|secure_redirects|
accept_source_route|tcp_synookies" /etc/sysctl.conf
net.ipv4.conf.all.accept_redirects=0
#net.ipv6.conf.all.accept_redirects = 0
```

```
net.ipv4.conf.all.secure_redirects=0
#net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.all.accept_source_route=0
#net.ipv6.conf.all.accept_source_route = 0
net.ipv4.conf.default.send_redirects=0
net.ipv4.conf.default.accept_redirects=0
net.ipv4.conf.default.secure_redirects=0
net.ipv4.conf.default.accept_source_route=0
adm_ivanov@p-fs-msk-03:~$ sysctl net.ipv4.tcp_synookies
sysctl: cannot stat /proc/sys/net/ipv4/tcp_synookies: No such file or directory
adm_ivanov@p-fs-msk-03:~$ sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0
adm_ivanov@p-fs-msk-03:~$ grep pam_wheel /etc/pam.d/su
auth       required   pam_wheel.so use uid group=admins
# auth       sufficient pam_wheel.so trust
# auth       required   pam_wheel.so deny group=nosu
adm_ivanov@p-fs-msk-03:~$ systemctl status rsyslog 2>/dev/null || echo "rsyslog not found"
rsyslog not found
adm_ivanov@p-fs-msk-03:~$ s -la /var/log/
-bash: s: command not found
adm_ivanov@p-fs-msk-03:~$ ls -la /var/log/
total 420
drwxrwxr-x 1 root   syslog   4096 Aug 29 12:24 .
drwxr-xr-x 1 root   root     4096 Aug 10 14:58 ..
-rw-r--r-- 1 root   root    11002 Aug 29 12:21 alternatives.log
drwxr-xr-x 1 root   root     4096 Aug 29 12:21 apt
-rw-r--r-- 1 syslog adm      1295 Aug 29 13:00 auth.log
-rw-r--r-- 1 root   root    64549 Aug 10 14:50 bootstrap.log
-rw-rw---- 1 root   utmp        0 Aug 10 14:50 btmp
-rw-r--r-- 1 root   root   244874 Aug 29 12:22 dpkg.log
-rw-r--r-- 1 root   root    32064 Aug 29 12:22 faillog
drwxr-xr-x 4 root   root     4096 Aug 29 12:21 kaspersky
-rw-r----- 1 syslog adm       252 Aug 29 12:24 kern.log
-rw-rw-r-- 1 root   utmp   292584 Aug 29 13:00 lastlog
drwxr-x--- 1 root   adm      4096 Aug 29 12:22 samba
drwx------ 3 root   root     4096 Aug 29 12:22 sudo-io
-rw------- 1 root   root      348 Aug 29 13:00 sudo.log
drwxr-xr-x 1 root   root     4096 Aug 29 12:22 supervisor
-rw-r--r-- 1 syslog adm      3130 Aug 29 12:24 syslog
-rw-rw-r-- 1 root   utmp      384 Aug 29 13:00 wtmp
adm_ivanov@p-fs-msk-03:~$ hostname
p-fs-msk-03
adm_ivanov@p-fs-msk-03:~$ ls -la /srv/files
ls: cannot access '/srv/files': No such file or directory
adm_ivanov@p-fs-msk-03:~$ getfacl /srv/files
getfacl: /srv/files: No such file or directory
adm_ivanov@p-fs-msk-03:~$ groups adm_ivanov
adm_ivanov : adm_ivanov admins
```

adm_ivanov@p-fs-msk-03:~$

Чек-лист для ХОСТ 4 – DNS-сервер (p-dns-spb-02)

IP: `192.168.10.70` (если не подключается, попробуйте `192.168.50.70` согласно отчёту пентеста)
Подключение: `ssh adm_ivanov@192.168.10.70`

Выполните те же общие проверки, что и для предыдущих хостов (команды 1–11), **плюс специфичные для DNS**:

Общие проверки (повторить как на хосте 3)

1. **Парольная политика**
   bash

grep ENCRYPT_METHOD /etc/login.defs
grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs

Запрет общих учётных записей

bash
cat /etc/passwd | grep -E "admin|user|test|shared"
last root

Минимальные привилегии (группы sudo)

bash
getent group sudo
getent group wheel
sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"

Логирование sudo

bash
sudo grep logfile /etc/sudoers
ls -la /var/log/sudo.log

Запрет root login по SSH

bash
grep -i permitrootlogin /etc/ssh/sshd_config
sshd -T | grep permitrootlogin

Сетевая фильтрация

bash
iptables -L -n -v
firewall-cmd --list-all 2>/dev/null

ufw status 2>/dev/null

Отключение неиспользуемых сервисов

bash
systemctl list-unit-files --state=enabled 2>/dev/null || echo "systemctl not available"
systemctl list-units --type=service --state=running 2>/dev/null || echo "systemctl not available"

Sysctl

bash
grep -E "send_redirects|accept_redirects|secure_redirects|accept_source_route|tcp_synookies" /etc/sysctl.conf
sysctl net.ipv4.tcp_synookies
sysctl net.ipv4.conf.all.accept_redirects

Ограничение su

bash
grep pam_wheel /etc/pam.d/su

Логирование событий

bash
systemctl status rsyslog 2>/dev/null || echo "rsyslog not found"
ls -la /var/log/auth.log   # или secure
ls -la /var/log/syslog   # или messages

Именование

bash
hostname
hostnamectl

Специфичные для DNS-сервера (из отчёта пентеста)

12. **Проверить статус DNS-сервиса (named/bind)**
    bash

Проверить, запущен ли сервис
ps aux | grep named
или
systemctl status named 2>/dev/null || systemctl status bind9 2>/dev/null || echo "named not found"

Проверить, отвечает ли на запросы (если запущен)

bash
Проверить открытые порты DNS (53)
ss -tulnp | grep :53

или
netstat -tulnp | grep :53

Проверить конфигурацию named (если есть) - НЕТ

bash
Проверить файл конфигурации
cat /etc/named.conf 2>/dev/null || cat /etc/bind/named.conf 2>/dev/null || echo "named.conf not found"
Проверить наличие зон
Проверить параметры recursion no, allow-query ограничен

Если сервис не используется, убедиться, что он отключён

bash
Если запущен, то остановить
sudo systemctl stop named && sudo systemctl disable named

adm_ivanov@p-dns-spb-02:~$ grep ENCRYPT_METHOD /etc/login.defs
This variable is deprecated. You should use ENCRYPT_METHOD.
ENCRYPT_METHOD SHA512
Only used if ENCRYPT_METHOD is set to SHA256 or SHA512.
adm_ivanov@p-dns-spb-02:~$ grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs
PASS_MAX_DAYS   Maximum number of days a password may be used.
PASS_WARN_AGE   Number of days warning given before a password expires.
PASS_MAX_DAYS 60
PASS_WARN_AGE 14
#PASS_MIN_LEN
PASS_MIN_LEN 17
adm_ivanov@p-dns-spb-02:~$ cat /etc/passwd | grep -E "admin|user|test|shared"
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
adm_ivanov@p-dns-spb-02:~$ last root

wtmp begins Sat Aug 29 13:05:00 2026
adm_ivanov@p-dns-spb-02:~$ getent group sudo
sudo:x:27:
adm_ivanov@p-dns-spb-02:~$ sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"
Defaults   env_reset
Defaults   mail_badpass
Defaults   secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults   use_pty
root   ALL=(ALL:ALL) ALL
%admin ALL=(ALL) ALL
%sudo   ALL=(ALL:ALL) ALL
@includedir /etc/sudoers.d
Defaults   logfile="/var/log/sudo.log"

```
Defaults    log_input, log_output
adm_ivanov@p-dns-spb-02:~$ sudo grep logfile /etc/sudoers
Defaults    logfile="/var/log/sudo.log"
adm_ivanov@p-dns-spb-02:~$ ls -la /var/log/sudo.log
-rw------- 1 root root 348 Aug 29 13:05 /var/log/sudo.log
adm_ivanov@p-dns-spb-02:~$ grep -i permitrootlogin /etc/ssh/sshd_config
PermitRootLogin no
# the setting of "PermitRootLogin without-password".
adm_ivanov@p-dns-spb-02:~$ sshd -T | grep permitrootlogin
sshd: no hostkeys available -- exiting.
adm_ivanov@p-dns-spb-02:~$ iptables -L -n -v
-bash: iptables: command not found
adm_ivanov@p-dns-spb-02:~$ firewall-cmd --list-all 2>/dev/null
adm_ivanov@p-dns-spb-02:~$ ufw status 2>/dev/null
adm_ivanov@p-dns-spb-02:~$ systemctl list-unit-files --state=enabled 2>/dev/null || echo
"systemctl not available"
systemctl not available
adm_ivanov@p-dns-spb-02:~$ sudo systemctl list-unit-files --state=enabled 2>/dev/null ||
echo "systemctl not available"
systemctl not available
adm_ivanov@p-dns-spb-02:~$ grep -E "send_redirects|accept_redirects|secure_redirects|
accept_source_route|tcp_synookies" /etc/sysctl.conf
#net.ipv4.conf.all.accept_redirects = 0
#net.ipv6.conf.all.accept_redirects = 0
# net.ipv4.conf.all.secure_redirects = 1
#net.ipv4.conf.all.send_redirects = 0
#net.ipv4.conf.all.accept_source_route = 0
#net.ipv6.conf.all.accept_source_route = 0
adm_ivanov@p-dns-spb-02:~$ sysctl net.ipv4.tcp_synookies
sysctl: cannot stat /proc/sys/net/ipv4/tcp_synookies: No such file or directory
adm_ivanov@p-dns-spb-02:~$ sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0
adm_ivanov@p-dns-spb-02:~$ grep pam_wheel /etc/pam.d/su
# auth       required   pam_wheel.so
# auth       sufficient pam_wheel.so trust
# auth       required   pam_wheel.so deny group=nosu
adm_ivanov@p-dns-spb-02:~$ ls -la /var/log/
total 380
drwxrwxr-x 1 root   syslog   4096 Aug 29 12:21 .
drwxr-xr-x 1 root   root     4096 Aug 10 14:58 ..
-rw-r--r-- 1 root   root    10344 Aug 29 12:20 alternatives.log
drwxr-xr-x 1 root   root     4096 Aug 29 12:19 apt
-rw-r--r-- 1 syslog adm      1608 Aug 29 13:05 auth.log
-rw-r--r-- 1 root   root    64549 Aug 10 14:50 bootstrap.log
-rw-rw---- 1 root   utmp        0 Aug 10 14:50 btmp
-rw-r--r-- 1 root   root   218015 Aug 29 12:20 dpkg.log
-rw-r--r-- 1 root   root    32032 Aug 29 12:21 faillog
-rw-rw-r-- 1 root   utmp   292292 Aug 29 13:05 lastlog
```

```
drwx------ 3 root   root     4096 Aug 29 12:21 sudo-io
-rw------- 1 root   root      348 Aug 29 13:05 sudo.log
-rw-r--r-- 1 syslog adm       515 Aug 29 12:21 syslog
-rw-rw-r-- 1 root   utmp      384 Aug 29 13:05 wtmp
adm_ivanov@p-dns-spb-02:~$ hostname
p-dns-spb-02
adm_ivanov@p-dns-spb-02:~$ ps aux | grep named
adm_iva+     139  0.0  0.0   4024  2048 pts/0    S+   13:07   0:00 grep --color=auto named
adm_ivanov@p-dns-spb-02:~$ ss -tulnp | grep :53
udp   UNCONN 0      0                *:53              *:*
tcp   LISTEN 0      4096             *:53              *:*
adm_ivanov@p-dns-spb-02:~$ cat /etc/named.conf 2>/dev/null || cat /etc/bind/named.conf
2>/dev/null || echo "named.conf not found"
named.conf not found
```

Чек-лист для ХОСТ 5 – АРМ Петрова (petrov-linux-001)

IP: `192.168.1.10`
Подключение: `ssh adm_ivanov@192.168.1.10`

Выполните **все общие проверки**, как для предыдущих хостов (команды 1–11), **плюс специфичные для АРМ**:

Общие проверки (повторить)

1. **Парольная политика**
   bash

```
grep ENCRYPT_METHOD /etc/login.defs
grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs
```

Запрет общих учётных записей

```
bash
cat /etc/passwd | grep -E "admin|user|test|shared"
last root
```

Минимальные привилегии (группы sudo)

```
bash
getent group sudo
getent group wheel
groups petrov   # проверить, не входит ли Петров в sudo-группы
sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"
```

Логирование sudo

bash
sudo grep logfile /etc/sudoers
ls -la /var/log/sudo.log

Запрет root login по SSH

bash
grep -i permitrootlogin /etc/ssh/sshd_config
sshd -T | grep permitrootlogin

Сетевая фильтрация

bash
iptables -L -n -v
firewall-cmd --list-all 2>/dev/null
ufw status 2>/dev/null

Отключение неиспользуемых сервисов

bash
systemctl list-unit-files --state=enabled 2>/dev/null || echo "systemctl not available"
systemctl list-units --type=service --state=running 2>/dev/null || echo "systemctl not available"

Sysctl

bash
grep -E "send_redirects|accept_redirects|secure_redirects|accept_source_route|tcp_synookies" /etc/sysctl.conf
sysctl net.ipv4.tcp_synookies
sysctl net.ipv4.conf.all.accept_redirects

Ограничение su

bash
grep pam_wheel /etc/pam.d/su

Логирование событий

bash
systemctl status rsyslog 2>/dev/null || echo "rsyslog not found"
ls -la /var/log/auth.log   # или secure
ls -la /var/log/syslog   # или messages

Именование

bash
hostname
hostnamectl

Специфичные для АРМ (из отчёта пентеста)

12. **Проверить права пользователя (не должен быть в sudo)**
    bash

```
id petrov   # проверить группы
sudo -l -U petrov   # список доступных sudo-команд (если есть)

adm_ivanov@petrov-linux-001:~$ grep ENCRYPT_METHOD /etc/login.defs
# This variable is deprecated. You should use ENCRYPT_METHOD.
ENCRYPT_METHOD SHA512
# Only used if ENCRYPT_METHOD is set to SHA256 or SHA512.
adm_ivanov@petrov-linux-001:~$ grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|
PASS_WARN_AGE" /etc/login.defs
#    PASS_MAX_DAYS    Maximum number of days a password may be used.
#    PASS_WARN_AGE    Number of days warning given before a password expires.
PASS_MAX_DAYS 90
PASS_WARN_AGE 14
#PASS_MIN_LEN
PASS_MIN_LEN 12
adm_ivanov@petrov-linux-001:~$ cat /etc/passwd | grep -E "admin|user|test|shared"
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
test:x:1002:1003::/home/test:/bin/bash
adm_ivanov@petrov-linux-001:~$ last root

wtmp begins Sat Aug 29 13:09:16 2026
adm_ivanov@petrov-linux-001:~$ getent group sudo
sudo:x:27:petrov
adm_ivanov@petrov-linux-001:~$ groups petrov
petrov : petrov sudo
adm_ivanov@petrov-linux-001:~$ sudo cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults    env_reset
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults    use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"
```

```
# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo    ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
adm_ivanov@petrov-linux-001:~$ sudo grep logfile /etc/sudoers
adm_ivanov@petrov-linux-001:~$ ls -la /var/log/sudo.log
ls: cannot access '/var/log/sudo.log': No such file or directory
adm_ivanov@petrov-linux-001:~$ grep -i permitrootlogin /etc/ssh/sshd_config
PermitRootLogin no
# the setting of "PermitRootLogin without-password".
adm_ivanov@petrov-linux-001:~$ sshd -T | grep permitrootlogin
sshd: no hostkeys available -- exiting.
adm_ivanov@petrov-linux-001:~$ iptables -L -n -v
-bash: iptables: command not found
```

```
adm_ivanov@petrov-linux-001:~$ systemctl list-unit-files --state=enabled 2>/dev/null || echo
"systemctl not available"
UNIT FILE                  STATE   VENDOR PRESET
cron.service               enabled enabled
dmesg.service                enabled enabled
e2scrub_reap.service         enabled enabled
getty@.service               enabled enabled
networkd-dispatcher.service enabled enabled
rsyslog.service              enabled enabled
ssh.service                 enabled enabled
systemd-pstore.service       enabled enabled
systemd-resolved.service     enabled enabled
systemd-timesyncd.service    enabled enabled
vsftpd.service               enabled enabled
remote-fs.target             enabled enabled
apt-daily-upgrade.timer      enabled enabled
apt-daily.timer              enabled enabled
dpkg-db-backup.timer         enabled enabled
e2scrub_all.timer            enabled enabled
fstrim.timer                enabled enabled
logrotate.timer              enabled enabled
motd-news.timer              enabled enabled

19 unit files listed.
adm_ivanov@petrov-linux-001:~$ systemctl list-units --type=service --state=running
2>/dev/null || echo "systemctl not available"
systemctl not available
adm_ivanov@petrov-linux-001:~$ grep -E "send_redirects|accept_redirects|
secure_redirects|accept_source_route|tcp_synookies" /etc/sysctl.conf
#net.ipv4.conf.all.accept_redirects = 0
#net.ipv6.conf.all.accept_redirects = 0
# net.ipv4.conf.all.secure_redirects = 1
#net.ipv4.conf.all.send_redirects = 0
#net.ipv4.conf.all.accept_source_route = 0
#net.ipv6.conf.all.accept_source_route = 0
adm_ivanov@petrov-linux-001:~$ sysctl net.ipv4.tcp_synookies
sysctl: cannot stat /proc/sys/net/ipv4/tcp_synookies: No such file or directory
adm_ivanov@petrov-linux-001:~$ sysctl net.ipv4.conf.all.accept_redirects
net.ipv4.conf.all.accept_redirects = 0
adm_ivanov@petrov-linux-001:~$ grep pam_wheel /etc/pam.d/su
auth       required   pam_wheel.so use uid group=admins
# auth       sufficient pam_wheel.so trust
# auth       required   pam_wheel.so deny group=nosu
adm_ivanov@petrov-linux-001:~$ systemctl status rsyslog 2>/dev/null || echo "rsyslog not
found"
rsyslog not found
adm_ivanov@petrov-linux-001:~$ ls -la /var/log/auth.log
-rw-r--r-- 1 syslog adm 1456 Aug 29 13:10 /var/log/auth.log
```

adm_ivanov@petrov-linux-001:~$ ls -la /var/log/syslog
-rw-r--r-- 1 syslog adm 3349 Aug 29 12:24 /var/log/syslog
adm_ivanov@petrov-linux-001:~$ hostname
petrov-linux-001
adm_ivanov@petrov-linux-001:~$ id petrov
uid=1001(petrov) gid=1002(petrov) groups=1002(petrov),27(sudo)
adm_ivanov@petrov-linux-001:~$ sudo -l -U petrov
Matching Defaults entries for petrov on petrov-linux-001:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/
sbin\:/bin\:/snap/bin, use_pty

User petrov may run the following commands on petrov-linux-001:
    (ALL : ALL) ALL
adm_ivanov@petrov-linux-001:~$

Чек-лист для ХОСТ 6 – АРМ Александрова (aleksandrov-ubuntu-01)

IP: `192.168.1.20`
Подключение: `ssh adm_ivanov@192.168.1.20`

Выполните те же команды, что и для АРМ Петрова (ХОСТ 5). Все проверки идентичны, только вместо `petrov` будет `aleksandrov`.

Команды для ХОСТ 6:

1. **Парольная политика**
   bash

grep ENCRYPT_METHOD /etc/login.defs
grep -E "PASS_MAX_DAYS|PASS_MIN_LEN|PASS_WARN_AGE" /etc/login.defs

Запрет общих учётных записей

bash
cat /etc/passwd | grep -E "admin|user|test|shared"
last root

Минимальные привилегии

bash
getent group sudo
getent group wheel
groups aleksandrov
sudo cat /etc/sudoers | grep -v "^#" | grep -v "^$"

Логирование sudo

bash
sudo grep logfile /etc/sudoers
ls -la /var/log/sudo.log

Запрет root login по SSH

bash
grep -i permitrootlogin /etc/ssh/sshd_config
sshd -T | grep permitrootlogin

Сетевая фильтрация

bash
iptables -L -n -v
firewall-cmd --list-all 2>/dev/null
ufw status 2>/dev/null

Отключение неиспользуемых сервисов

bash
systemctl list-unit-files --state=enabled 2>/dev/null || echo "systemctl not available"
systemctl list-units --type=service --state=running 2>/dev/null || echo "systemctl not available"

Sysctl

bash
grep -E "send_redirects|accept_redirects|secure_redirects|accept_source_route|tcp_synookies" /etc/sysctl.conf
sysctl net.ipv4.tcp_synookies
sysctl net.ipv4.conf.all.accept_redirects

Ограничение su

bash
grep pam_wheel /etc/pam.d/su

Логирование событий

bash
systemctl status rsyslog 2>/dev/null || echo "rsyslog not found"
ls -la /var/log/auth.log
ls -la /var/log/syslog

Именование

bash
hostname
hostnamectl

Проверка прав пользователя

bash
id aleksandrov
sudo -l -U aleksandrov

—--
ХОСТ 7
========== ОБЩИЕ ПРОВЕРКИ ==========

--- 1. Парольная политика ---
This variable is deprecated. You should use ENCRYPT_METHOD.
ENCRYPT_METHOD SHA512
Only used if ENCRYPT_METHOD is set to SHA256 or SHA512.
#	PASS_MAX_DAYS	Maximum number of days a password may be used.
#	PASS_WARN_AGE	Number of days warning given before a password expires.
PASS_MAX_DAYS 90
PASS_WARN_AGE 14
#PASS_MIN_LEN
PASS_MIN_LEN 12

--- 2. Общие учётные записи ---
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
Последние входы root:

wtmp begins Sat Aug 29 13:17:40 2026

--- 3. Группы привилегий ---
sudo:x:27:
Группа wheel не найдена
Содержимое /etc/sudoers (без комментариев):
Defaults	env_reset
Defaults	mail_badpass
Defaults	secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults	use_pty
root	ALL=(ALL:ALL) ALL
%admin ALL=(ALL) ALL
%sudo	ALL=(ALL:ALL) ALL
@includedir /etc/sudoers.d
Defaults	logfile="/var/log/sudo.log"
Defaults	log_input, log_output

--- 4. Логирование sudo ---
Defaults	logfile="/var/log/sudo.log"
-rw------- 1 root root 686 Aug 29 13:19 /var/log/sudo.log

--- 5. Root login SSH ---
PermitRootLogin yes
the setting of "PermitRootLogin without-password".
permitrootlogin yes

--- 6. Файрвол ---
Не удалось определить фаервол

--- 7. Включенные сервисы (первые 20) ---
UNIT FILE STATE VENDOR PRESET
cron.service enabled enabled
dmesg.service enabled enabled
e2scrub_reap.service enabled enabled
getty@.service enabled enabled
networkd-dispatcher.service enabled enabled
rsyslog.service enabled enabled
ssh.service enabled enabled
systemd-pstore.service enabled enabled
systemd-resolved.service enabled enabled
systemd-timesyncd.service enabled enabled
remote-fs.target enabled enabled
apt-daily-upgrade.timer enabled enabled
apt-daily.timer enabled enabled
dpkg-db-backup.timer enabled enabled
e2scrub_all.timer enabled enabled
fstrim.timer enabled enabled
logrotate.timer enabled enabled
motd-news.timer enabled enabled

--- 8. sysctl параметры ---
net.ipv4.conf.default.send_redirects = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 1
net.ipv4.conf.all.secure_redirects = 1
net.ipv4.conf.default.secure_redirects = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 1
net.ipv4.tcp_synookies = (не задан)

--- 9. Ограничение su ---
auth required pam_wheel.so use uid group=admins

--- 10. Статус rsyslog и права логов ---
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
-rw-r--r-- 1 syslog adm 2795 Aug 29 13:19 /var/log/auth.log

-rw-r--r-- 1 syslog adm 1245 Aug 29 12:23 /var/log/syslog

--- 11. Имя хоста ---
ivanov-linux-003

--- Пользователи в группе sudo/wheel ---

========== РАБОЧАЯ СТАНЦИЯ ==========
Проверка пользователя ivanov (должен быть в admins):
ivanov : ivanov

========== ПРОВЕРКА ЗАВЕРШЕНА ==========